



BSC. (HONS) COMPUTER SCIENCE AND SOFTWARE ENGINEERING

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TripMate: A Webapp for collaborative planning and managing trips in Sri Lanka

Contextual Report

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Contents

List of Figures	ii
List of Tables	iii
1. Introduction.....	1
1.1 Project Background.....	1
1.1.1 Motivation.....	1
1.1.2 Research Problem	1
1.2 Aim and objectives of the project	2
1.3 Description of the artifact	2
1.4 Structure of the contextual report	3
2. Literature Review.....	4
2.1 Introduction.....	4
2.2 Key Points.....	5
2.2.1 AI-Powered Travel Planner.....	5
2.2.2 The role of Large Language Models (LLMs) in collaborative travel planning.	6
2.2.3 Group tour trip recommender model for travel systems.....	7
2.2.4 Seamless Group Trip Planning.....	8
2.2.5 Machine learning algorithms for constructions cost prediction: A systematic review	9
2.2.6 Group recommender systems for tourism: how does personality predict preferences for attractions, travel motivations, preferences and concerns	10
2.3 Similar Systems	10
2.3.1 TripIt	10
2.3.2 RoadTrippers.....	13
2.3.3 SaveTrip.....	16
2.4 Research Gap	18
2.5 Market Research	20
2.5.1 Data Gathering	20
2.5.2 Requirement Analysis	20
2.5.3 Conclusion	27
3. Project Plan	28
3.1 Overview of the project	28

3.2	Gantt Chart.....	28
3.3	Work Breakdown Structure.....	30
4.	Planning the Artifact	31
4.1	Methodology	31
4.2	Planning the Artifact	32
4.3	Requirement gathering and Analysis	32
4.4	Design and Implementation	35
4.5	Testing and Evaluation.....	35
4.6	Risks and Risk Mitigation Techniques.....	35
	References.....	36
	Appendix A – Questionnaire.....	37
	Appendix B – Work Breakdown Structure	41

List of Figures

Figure 1 : TripIt Homepage 11

Figure 2 : Functions of TripIt 01.....	11
Figure 3 : Functions of TripIt 02.....	12
Figure 4 : Functions of TripIt 03.....	12
Figure 5 : Homepage of RoadTrippers	13
Figure 6 : Functions of RoadTrippers 01	14
Figure 7 : Functions of RoadTrippers 02	14
Figure 8 : Functions of RoadTrippers 03	15
Figure 9 : SaveTrip Homepage	16
Figure 10 : Functions of SaveTrip 01	17
Figure 11: Functions of SaveTrip 02	17
Figure 12 : Survey questionnaire, what is your age group?.....	20
Figure 13 : Survey questionnaire, how often do you travel in Sri Lanka?	21
Figure 14 : Survey questionnaire, how do you usually plan trips?.....	21
Figure 15 : Survey questionnaire, what challenges do you face when planning group trips?.22	
Figure 16 : Survey questionnaire, have you ever used a platform specifically designed for group trip planning?.....	23
Figure 17 : Survey questionnaire, how useful would a web app be that allows you to plan trips collaboratively with friends?	23
Figure 18 : Survey questionnaire how important are following features to you?.....	24
Figure 19 : Survey questionnaire, would you use a feature where the system suggests travel destinations based on your	25

Figure 20 : Survey questionnaire, how valuable is an automated cost estimation tool when planning a trip?	25
Figure 21: Survey questionnaire, what is most important for you when using a trip planning app?	26
Figure 22 : TripMate Gantt Chart	29
Figure 23 : WBS	30

List of Tables

Table 1: Comparison between existing systems and TripMate.....	19
Table 2 : Advantages and Disadvantages of Agile methodology.....	31
Table 3 : Hardware Requirements needed for TripMate.....	32
Table 4 : Software Requirements for TripMate	33
Table 5 : Functional Requirements of TripMate	33
Table 6 : Non-functional Requirements of TripMate.....	34

1. Introduction

The process of planning for a trip has become more and more technologically reliant with the advent of web-based trip planners and social networking sites. Although current web-based solutions offer access to maps, weather updates, and information about the place of visit and user-uploaded media contents, the process of planning for a trip and group trip planning specifically remains disorganized due to the lack of structured collaboration support. Group travel involves co-ordination of multiple people with access to group travel information and ability to collectively assess and arrive at a final travel solution. However, present-day online solutions for trip planning continue to be user-centric and lack an inherent workflow process for group collaboration and building group consensus towards arriving at group preferences. TripMate remedies this shortcoming by articulating and designing an interactive web-based system that supports a defined and transparent process for group trip planning.

1.1 Project Background

With the growth of technologies people have changed the way they travel with significant influence by digital platforms on searching for destinations, determining routes to take, and figuring out the travelling conditions. Most of the tools are focused mainly on solo trip planning or the activities after booking, with less to no support for real-time collaboration and shared decision making, causing group trip planning to rely on the use of messaging apps for communication, separate map services, and websites as information platforms. Therefore, creating a clear need for a single unified platform that allows users for collaboration and trip management.

1.1.1 Motivation

The motivation for the TripMate project has arisen by the problems encountered in the real world while planning group travel activities. When there is more than one individual involved, choices about destinations, routes, accommodation, and cost estimations tend to be lengthy and ineffective. Lack of a centralized collaborative tool creates confusion, lack of clarity and decreases user satisfaction. TripMate is driven by the desire to create a well-organized and clear planning space in which everyone in a group can contribute. The system will streamline the planning process and make the whole travel experience more enjoyable.

1.1.2 Research Problem

The major problem that the proposed project aims to solve is the absence of reliable collaboration in existing travel planning software. Currently, all existing systems are designed for individual use only and lack the following important aspects: group collaboration in a real-time context, voting or consensus-building processes, and easy access to relevant travel-related information in a timely context. Moreover, they do not provide assistance in

estimating costs either. Users are required to perform the calculations on their own, which causes complexity and creates ambiguity in group travel planning. To make it worse, the unavailability of NLP interfaces also makes the user interaction experience less natural. This creates more challenges in the context of group preferences and taking collective inputs and relevant insights from scattered travel-related information. This proposed project discusses the potential benefits of using a web-based group collaboration tool designed in a way to include NLP interfaces and group-related relevant travel information in group travel planning.

1.2 Aim and objectives of the project

To facilitate the efficient planning, organization, and management of trips by a group of people by allowing them to make group decisions and providing travel information.

This project has the following objectives:

1. To develop an interactive web interface which will facilitate the process of sharing trip plans.
2. To apply a secure user authentication and access control based on actively contributing group(s).
3. To allow group members to propose and vote on the destinations, routes and accommodation together.
4. To map, weather forecast, and travel video API to offer contextual travel data.
5. To create a cost estimation tool, which computes the overall and the individual costs of traveling.
6. To provide data integrity, security and scalability in the system.

1.3 Description of the artifact

TripMate is a webapp that assists in the planning of trips by groups. The application gives the users the ability to plan their trips, share with participants, and organize travel plans together. Members of groups are able to share ideas, find information about the destination, plot routes on interactive maps, and undergo organized voting. The system also can offer approximate costs of travel using parameters related to the trips to make decisions based on the information in the planning stage.

The TripMate project scope comprises collaborative trip creation, group decision-making as well as the integration with related APIs and cost-estimation prediction. The system concentrates only on the process of planning and coordination of travel. Other functions like booking, purchase of tickets and all other payment processes will be beyond the scope of this project. This application is designed as a web-based application that can be extended to the future.

The specified issue is solved through the elaboration of a centralized collaborative platform uniting the communication, the decision-making process, and travel-related

information. TripMate allows organizations to support decision-making about travel through shared planning, structured voting, and contextual data through external API connections, and cost estimation, making informed and transparent decisions in the same system.

1.4 Structure of the contextual report

The contextual report is organized into four key sections.

- Section 1 provides an introduction of the project including the background, motivation, research problem, aim, objectives, scope and a description of the artefact that is proposed.
- Section 2 provides a Literature Review and Market Research. The literature review considers academic literature, technologies, and other similar systems in relation to collaborative travel planning and digital solutions to support it. The market research section provides a work plan of the study, data gathering procedures, findings analysis, and causes results discussion to determine the existing gaps and user requirements that are applicable to the proposed system.
- Section 3 describes the Project Plan, including the general approach to the project, work breakdown structure, milestones and time schedule, with the help of planning tools.
- Section 4 is focused on Planning the Artefact, which explains the choice of the development methodology, the process of gathering and analyzing requirements, system design considerations, the way of implementation, the way of testing and evaluating the system, and the way of eliminating risks.

2. Literature Review

This section summarizes the available literature and solutions used in the industry in relation to collaborative travel planning systems. It reviews existing research on AI-based travel planners, group recommender systems, large language model applied in planning, cost estimation methods, and personality-based recommendation methods and analyses similar commercial products. This review aims at identifying existing capabilities, limitations, and emerging trends in the field and to set the informed base of the suggested TripMate system by outlining a gap in the research and practice that the project is aimed to fill.

2.1 Introduction

The digital travel planning problem domain that this literature review will touch on is the group-based and collaborative travel planning systems. Although the current web technologies and artificial intelligence have enhanced access to travel-related information, the current solutions still face issues with structuring collaborative work, group decision-making, and transparent planning processes.

This review explores the attempts of the current research and systems to address these challenges and where the limitations still exist. Academic research and available systems associated with AI-based travel planning, large language models in planning, group-based recommender systems, machine learning-based optimization, and cost prediction methods can be considered as the scope of this literature review.

Besides that, the review takes into account commercial travel planning applications in order to learn how practical systems aid or restrict collaborative planning. The primary areas of this review are AI based trip planning and itinerary optimization, the use of large language models in collaborative travel planning, group tour recommender models to support personalized travel systems, group coordination and decision facilitation with the help of AI, machine learning methods that are applicable to cost prediction, and personality-sensitive group recommender systems.

Moreover, some of the existing systems that resemble the proposed one, e.g., TripIt, RoadTrippers, and SaveTrip are inspected to assess their functionality and teamwork feature. The major aspects covered in the review are those which find out the strengths and weaknesses of the current methods especially when it comes to collaboration, real-time engagement, preference bargaining, cost estimation and user participation in group decision making. The insights create a baseline to determine the gaps within the existing systems and the reason why collaborative web-based travel planning solution is necessary.

2.2 Key Points

2.2.1 AI-Powered Travel Planner

The Application of Artificial Intelligence (AI) has been used in contemporary travel planning in the creation of an AI-driven trip optimization system. This paper demonstrates the shortcomings of the conventional travel planning methods which generally force users to individually gather and consolidate data on numerous independent websites to create disjointed and unproductive travel plans. In order to overcome these shortcomings, the suggested system will combine methods of artificial intelligence and natural language processing to build individual travel guides based on the choices of the user, length of the travel, interests toward a specific destination, and contextual elements like weather conditions. The study compares the current platforms, such as Google Travel and TripHobo, and finds lack of flexibility, personalization, and adaptability. Also, it uses machine learning algorithms like collaborative, content-based, and genetic algorithms to optimize itinerary generation, including systems of unreliability of reviews detectors. Despite the fact that the system has proven to be more efficient and customizable when it comes to planning separate trips, the research specifically aims at solving problems of optimization in a single-user mode and offers minimal assistance in group decision-making. This constraint shows a shortcoming of existing travel planning systems and supports the necessity of collaborative web-based technology of coordinated planning and group decision-making. (Dhote *et al.*, 2025).

An AI-based trip planner system that endeavors to enhance travel planning efficiency and personalization by combining insights from artificial intelligence, machine learning, and data analysis techniques and methods. This examination reveals some of the most crucial shortcomings of typical travel planning solutions, which require users to manually collect relevant travel data from respective sourcing platforms, causing fragmented travel plans and considerable decision-making efforts and time. In order to address and rectify these and related travel planning issues, this proposed system can produce highly personalized travel schedules depending on the preferences, travel behavior, size of traveling parties, and budget constraints of relevant parties or individuals, besides adopting relevant real-time context information including weather conditions and travel setbacks or inconveniences. This proposed system utilizes machine learning algorithms like collaborative filtering optimization and related models for providing suggestions to travel parties regarding respective destinations, activities, and accommodations, besides making dynamic and real-time changes to travel plans depending on related changes and events, and experimental tests and validation of this proposed system demonstrate remarkable levels of user satisfaction regarding travel planning and relevant cost optimization, besides showing considerable improvements in travel planning efficiency and relevant cost reduction and savings, including important reductions in travel and related planning costs (Dr. Anitha S. *et al.*, 2025).

2.2.2 The role of Large Language Models (LLMs) in collaborative travel planning

The paper discusses how large language models can assist in collaborative travel planning the role of generative AI (particularly ChatGPT) in group-based information-seeking and decision-making. The research design is a within-subject experimental design, whereby the participant pairs intended to plan a short trip under two conditions, with one being the use of a traditional search engine and the other being the use of a search engine and ChatGPT. The results suggest that inclusion of ChatGPT did not significantly affect the overall duration of the time spent by the users on the task and the level of user satisfaction but positively affected the quality of the travel plans generated and the level of completeness. Participants who utilized ChatGPT were more efficient in generating destination ideas and generated plans that were more in line with task requirements especially in the initial ideation and planning phases. But the paper also depicts the limitations of generative AI in the work involving real time information as in the case of accommodation and transport planning where a traditional search engine proved to be more efficient. The study finds that large language models are effective as an auxiliary resource in collaborative travel planning, complementing idea generation and information organizing and not completely substituting traditional search techniques and human computations (Kumamoto and Joho, 2025).

Travel planning is a complex activity involving the ordering of tasks at points of interest (POIs) when faced with the challenges of the real world, and the maximization of utility. The traditional methods used in computing travel plans are Automated Planning (AP) and Mixed-Integer Programming, which ensure strong satisfaction of constraints but entail the "tedious and time-consuming" process of manual problem formulation. Although recent advances in the use of the Large Language Models (LLMs) such as GPT-4 are shown to be useful because of its large domain knowledge, it has been found to be a "bad reasoner" in problems requiring sequential decision-making. Research shows that it often neglects to create valid plans to satisfy hard constraints like travel time and stay time at each POI.

To overcome these inadequacies, the authors have introduced TRIP-PAL (Travel Itinerary Planning with Planners and Language models), a hybrid system that leverages the semantic strengths of LLMs and the logical strengths of automated planners. This framework utilizes the LLM as a translation interface that parses the input POIs and their popularity-based utilities and temporal needs into the planning system. This information is converted into the Planning Domain Definition Language (PDDL) problem.

The value of this research lies in addressing oversubscription planning problems, when the number of desired goals (POIs) is more than the available time budget. To counter this problem in TRIP-PAL, the goal points are considered to be "soft goals." A cost-minimization function is developed in this approach to maximize customer satisfaction.

“Experimental assessments of TRIP-PAL vs GPT-4 demonstrate a marked disparity in reliability for:

- Plan Validity: The performance of GPT-4 has been poor with only 14 valid plans out of 100 in normal situations and regularly violating visit/travel times.
- Optimality: When the hard constraints are disregarded to give LLM a better chance, TRIP-PAL outperformed it in terms of optimized routes in 79% of the test scenarios.
- Scalability: While the number of POIs rose, GPT-4’s rate of success dropped dramatically, whereas in TRIP-PAL, the result was always valid and/or optimal, although with a larger running time for difficult tasks (up to additional 800 seconds for 18 POIs).

Ultimately, it is proven through research that although LLMs are very useful for information retrieval, they are still unsuitable for replacing specialized solvers. It is evident from TRIP-PAL that a hybrid model presents an “end-to-end solution to provide sound, constraint-compliant, and optimal routes.” (de la Rosa et al, 2024).

2.2.3 Group tour trip recommender model for travel systems

A new Group Tour Trip Recommender Model (GTTRM) that can handle the challenges of the personalized travel planning of groups with a wide range of and possibly conflicting interests. The research determines the major drawbacks of the current group recommender systems, mainly, the use of fixed aggregation functions that do not accommodate the changing preferences throughout a trip. To solve these issues, the model suggested will use the Ant Colony Optimization (ACO) to dynamically optimize the itineraries and allow formation of subgroups in real-time when the schedules of group members conflict with each other. In contrast to the old ways of creating an itinerary that would suit the whole group, GTTRM prepares a balance between the satisfaction of the group and the satisfaction of the individual by the ability to separate and reunite the groups temporarily, depending on the activity, time, and social relationships. The model incorporates a more detailed constraints data model, both of the activity- and connexion-level constraints and follows a time-based approach to optimization, which considers the travel time, waiting time and multi-value points of interest. Real-world experimental tests indicate that GTTRM has an overwhelming success over currently existing state-of-the-art in terms of traveler satisfaction, efficiency in optimization and flexibility. In general, the research has a significant contribution to group travel recommender systems because it proposes a dynamic, scalable, and highly personalized optimization framework applicable to actual collaborative travel planning situations (Alatiyyah, 2025).

A group travel recommender system tackles all the messy parts of group decision-making head-on. Most current systems stick to helping solo travelers or just push out one-size-fits-all suggestions, which means anyone with different tastes often gets ignored. The team behind this new approach mixes Collaborative Filtering (CF) to pick out travel destinations and an Approximate Constraint Satisfaction (ACSP) process to actually handle what everyone in the group wants. First, the system looks for “neighbor groups”—other groups with similar travel histories—to get a shortlist of possible destinations. Next, it takes everyone’s individual needs (like budget, hotel types, flight preferences) and sets them up as group-level requirements. What’s clever here is the “indifference interval.” This built-in wiggle room makes sure the system stays flexible, because, let’s face it, most travelers are willing to compromise a bit. Last, there’s a post-filtering step that kicks out any travel packages that seriously clash with one person’s must-haves—so even the quieter folks don’t end up sidelined. The research team tested the system with 210 seasoned travelers and found it beats standard models on all the important stuff: relevance, novelty, and keeping users happy. Groups of five, for example, saw average dissatisfaction with flight prices drop by 93.4 points. The system holds up no matter the group size, too. All in all, this research gives travel agencies a practical, privacy-friendly way to suggest entire travel packages, flights, hotels, activities—that keep everyone in the group happy (Kim et al, 2024).

2.2.4 Seamless Group Trip Planning

“Trip Sage, an intelligent solution to simplify old-fashioned trip planning via AI and automation” Historically, travelers have had to piece together broken information from travel agents and a multitude of websites manually—something the authors describe as slow work, also not always personalized. To overcome such drawbacks, the researchers proposed a system that used the Gemini AI model to suggest personalized timetables based on user-provided meta information such as destination name, trip days, number of travelers and budget.

The software stack is implemented using a React based frontend and Node.js/Express backend with Firestore for data storage in the cloud that you can trust. Honeybites Dynamic incorporation of the Google Maps API enables immediate access to locations as well as navigation help and optimal route planning. Unlike ordinary planners that might be built on static databases with outdated information, Trip Sage sources real-time data to give users meaningful suggestions about what to see and do getting there. The authors highlight the platform’s potential to enhance user satisfaction by automatically planning junkets with low effort consumption and incorporating smarter budget optimization and cost estimation for different travel expenditures. The study finds that the modular and scalable architecture of AI Trip Planner represents a considerable improvement compared to manual approaches, delivering an optimal, personalized, non-stressful solution for the contemporary traveler. (Ravindra et al., 2025).

The researchers examine the issues surrounding group trip planning and suggest an offer of digitally mediated answer that connects the personal travel inspirations with joint decision-making. The research is on the issues of how travelers, especially digitally literate travelers, have difficulty balancing their individual preferences with group decisions when making arrangements at a distance. The authors explore the current gaps in collaborative travel planning tools via a design-led research method, pointing out drawbacks like poor communication, absence of individualized suggestions and ineffective group decision-making. The study conceptualizes the idea of an AI-assisted collaborative planning system where personal inspirations are shared in a common planning platform by integrating saved social media materials, to create a set of personalized but negotiable itineraries. Moreover, the paper presents the use of an AI chatbot as a facilitator to enable equal participation, balance of interests, and direct groups to make collective decisions without delegating the human agency. The results indicate that AI can be successfully used to complement collaborative travel planning through the means of personalization, reduced cognitive load, and the overall enhanced efficiency of group coordination, though it is also necessary to pay close attention to the question of transparency, ethical considerations, and user trust (Patel and Armen, 2024).

2.2.5 Machine learning algorithms for constructions cost prediction: A systematic review

The researchers present a qualitative review of how artificial intelligence gradually assuming a new role in construction management, with regards to the complexity of conceptual cost estimation. The authors note that manual and static software-based estimation techniques are becoming ineffective in managing the non-linear variables that abound in the contemporary civil engineering project, like roads, bridges, and buildings. As a result, the use of machine learning (ML) and deep learning (DL) algorithms has increased significantly, and the volume of research in this area is growing significantly from 2017 to 2021. Their discussion points out Artificial Neural Networks (ANN) and Support Vector Machines (SVM) as the two most common and successful models which frequently reach predictive accuracies of 90-100%. Nevertheless, the research identifies serious systemic challenges, such as the use of old or small datasets, metadata that is not publicly available to be validated by third parties, and the research bias, where the focus is placed on building and road projects, rather than tunnels, dams, and airports. To enhance the validity of future cost predictors, the authors want to suggest adopting single computing platforms and obliging them to release open-access datasets

(Ghadbhan Abed, Hasan and Zehawi, 2022).

2.2.6 Group recommender systems for tourism: how does personality predict preferences for attractions, travel motivations, preferences and concerns

Tourist behavior studies have turned to psychological profiling in increasing numbers to improve the accuracy of the recommender system (RS). The authors examined the role of the Big Five personality dimensions in forecasting different aspects of travel, such as attraction preference, motivations and concerns. According to their analysis that covered 1,035 respondents, all of the five dimensions are useful in predicting attraction decisions and travel issues, but only two of them, Neuroticism and Openness, have a significant role in predicting travel motives. In particular, Openness is positively associated with diverse motives including novelty, self-development and enjoyment of nature, which are indicative of fundamental motive of intellectual stimulation and variety. Conversely, Neuroticism seems to have the strongest effect on motives that touch on self-development and relatedness and may be one of the ways that the self can enhance self-consciousness. The results of these studies enable the development of personality-sensitive group recommender systems (GRS) to address the cold-start problem by modelling the user profiles automatically and addressing any incompatibility in preferences among heterogeneous travel groups (Alves et al, 2023).

2.3 Similar Systems

2.3.1 TripIt

TripIt is a popular online travel planning and itinerary management app that takes over the process of organizing the details of a traveler into a single organized itinerary. Users are able to forward their confirmation emails on flights, hotels, car rentals and activities to an email address, after which TripIt pulls out all pertinent details of the booking and unpacks it into an all-encompassing itinerary that can be accessed across devices and platforms. The system also enables the integration of calendar and sharing of itineraries and so plans could be synced with external calendar or shared with other travelers. TripIt is quite efficient in carrying out travel information aggregation and delivery of real-time updates and information but has little to no collaborative abilities: the system can share itinerary with other people but cannot have several people editing the itinerary information and jointly planning the itinerary detail in real-time. TripIt also does not have native budgeting, cost-splitting, and decision-making tools, which limit its applicability to large-scale collaborative group planning where a joint input and consensus are needed. As much as they are popular when it comes to individual and business travel organization, these restrictions underscore the need to have more dedicated collaborative planning systems that will cover the group travel dynamics in a comprehensive manner (TripIt, 2025).

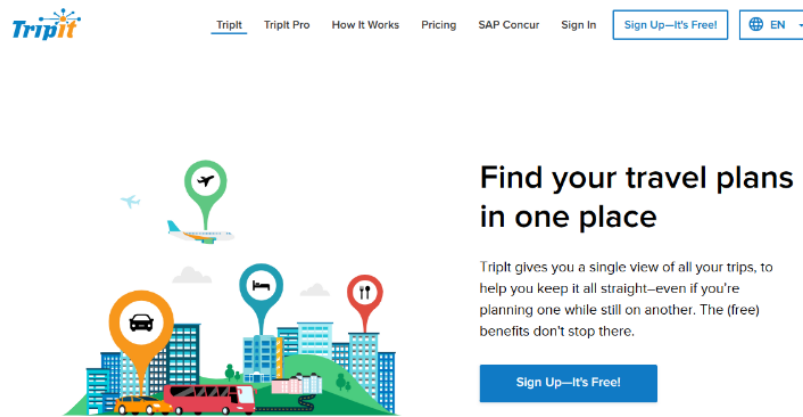


Figure 1 : TripIt Homepage

Figure 01 shows the homepage of the TripIt website.

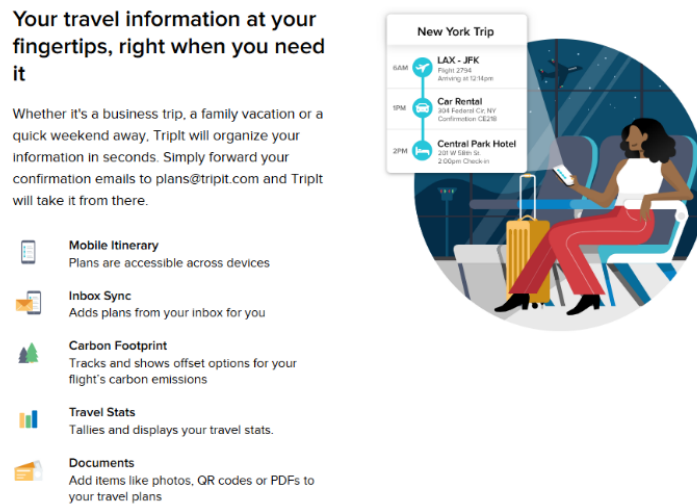


Figure 2 : Functions of TripIt 01

Figure 02 shows the functions of TripIt for managing the application including Mobile History, Inbox Sync, Carbon Footprint Tracking, Travel Stats, Document syncing.

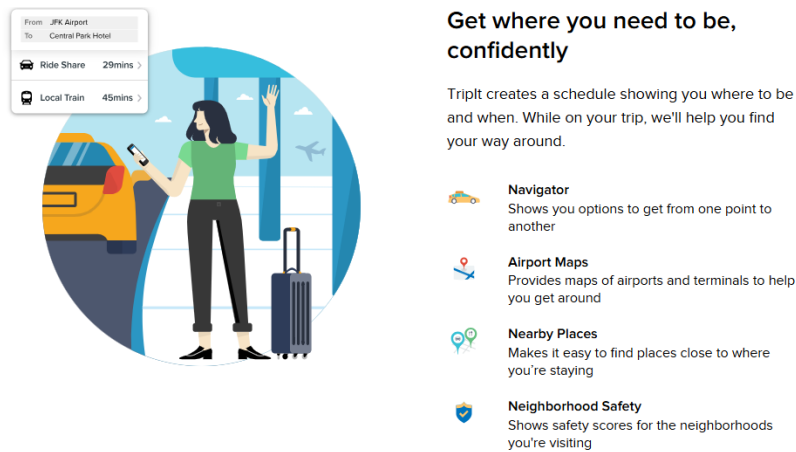
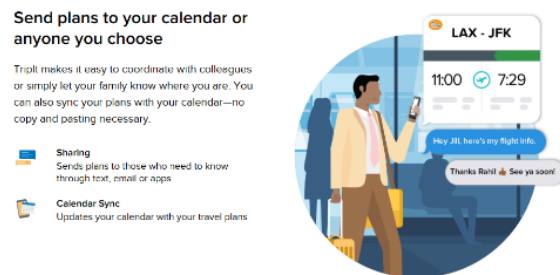


Figure 3 : Functions of TripIt 02

Figure 03 shows the functions of TripIt for navigation including Airport maps, Location of Nearby places, and Neighborhood safety scores.



Additional TripIt features

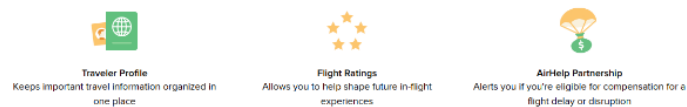


Figure 4 : Functions of TripIt 03

Figure 04 shows the other functions of TripIt including calendar syncing, keeping a travel profile, Flight ratings, and alerting for getting compensation for flight delays.

2.3.2 RoadTrippers

RoadTrippers is an online travel planning service that is meant to help its users find, plan and optimize road trips by incorporating itinerary creation, points of interest and route exploration into a single user interface. The system allows the traveler to enter start and endpoint destinations, explores places of interest and stops of the journey, and creates itineraries with costs and distance covering each day based on the car specifications and route requirements. It also has interactive maps, offline navigation, and collaborative trip editing and invited users can add and edit waypoints and share travel plans with companions. Although the RoadTrippers platform is mainly designed to facilitate the planning of individual trips, its collaboration capabilities allow multiple participants to work on the itinerary planning and review, though not in real time in a synchronous mode. Automated planning is also offered, with a tool called Autopilot, which is powered by AI and generates suggested routes and stops, depending on the preferences of the trip, as well as budgeting and routing assistance. RoadTrippers is an illustration of a new integrated travel planning tool that has a long discovery and collaboration feature, but it is oriented towards assisting individual people in road trip planning, and not high-level collaborative decision-making that is needed to find a common ground in planning a road trip (RoadTrippers, 2025).

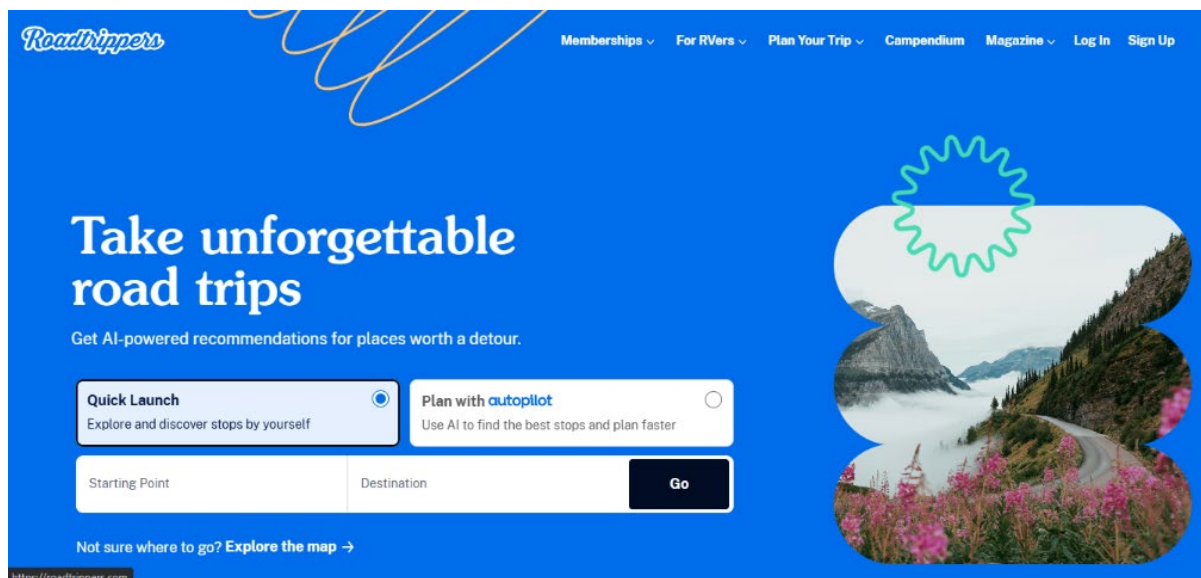


Figure 5 : Homepage of RoadTrippers

Figure 05 shows the homepage of RoadTrippers website.

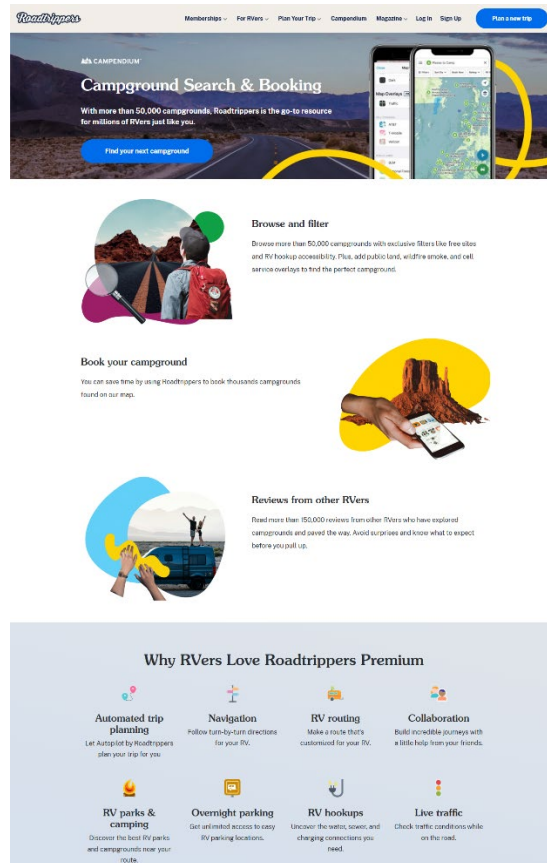


Figure 6 : Functions of RoadTrippers 01

Figure 06 shows the function of RoadTrippers campground booking, they provide services for solo travelers, and Recreational Vehicle (RV) campers.

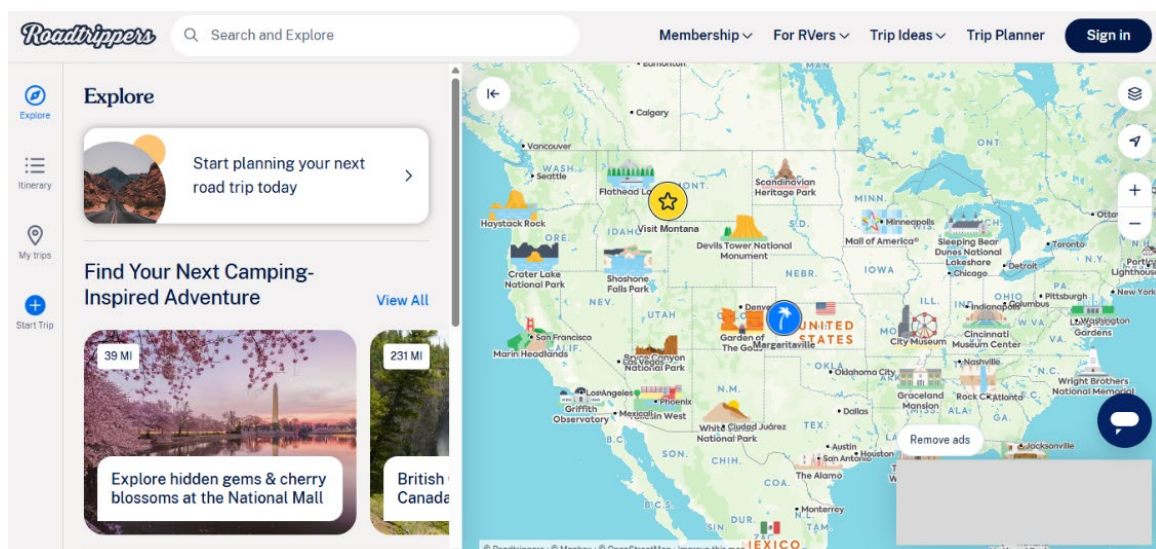


Figure 7 : Functions of RoadTrippers 02

Figure 07 shows the trip planning function of the RoadTrippers. It includes an interactive map and destination recommendations.

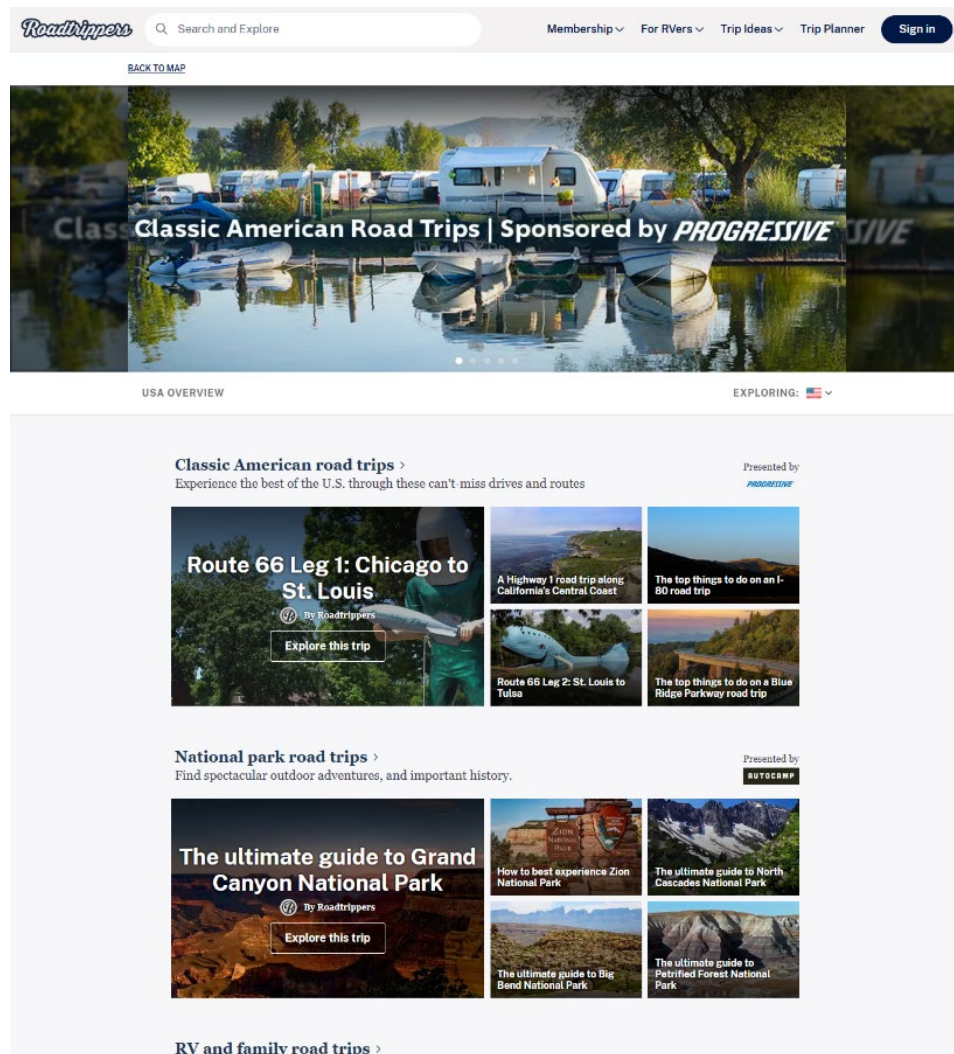


Figure 8 : Functions of RoadTrippers 03

Figure 08 shows the tour guides included in RoadTrippers. They include guides for solo trippers, Family tours, RV Campers and Themed tours.

2.3.3 SaveTrip

SaveTrip is an online travel planning software that allows users to plan the schedules and other financial aspects of a trip at a single interface. The platform enables travelers to create itineraries written down and translated into a chronological sequence and visualized on built-in maps, to aid route planning and navigation via third-party services like Google Maps. Besides itinerary management, SaveTrip also offers to track travel expenditures in various currencies to receive visualized expense reports and transform receipts into tabular expense entries. It also allows the users to add documents, photos, and checklists to make their travel notes rich and also export travel plans and expense report to standard file formats (csv, PDF) to review and save them offline. Even though SaveTrip bundles a number of planning and budgeting tools, it is more of a personal itinerary and expense management tool than a collaborative feature that allows sharing decisions made by multiple travelers. This weakness points to the fact that there is a void in the current range of travel-planning applications, which supports the development of systems that are more appropriate in terms of real-time and interactive group work (SaveTrip, 2025).

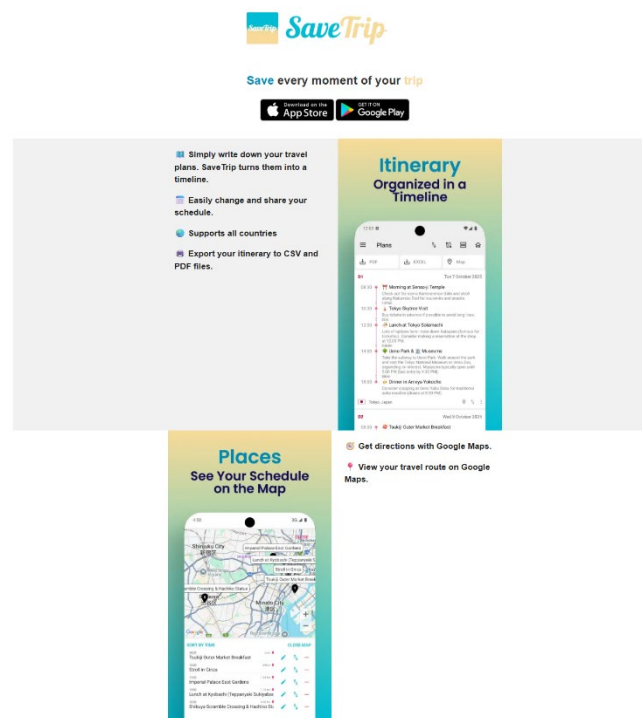


Figure 9 : SaveTrip Homepage

Figure 09 shows the homepage of SaveTrip including where you can download the mobile application and some functionalities including itinerary organizing and scheduling.



Figure 10 : Functions of SaveTrip 01

Figure 10 shows the functions of SaveTrip Expense tracking, Expense report generation and inbuilt Note taking application.

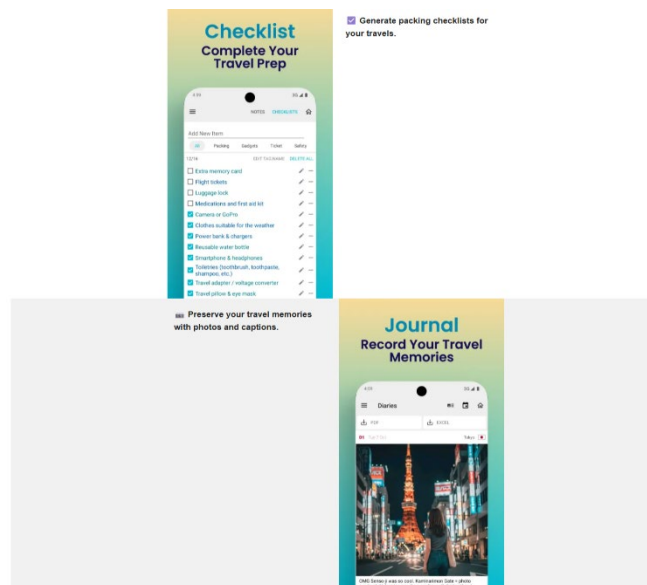


Figure 11: Functions of SaveTrip 02

Figure 11 shows the other functions of SaveTrip which includes creating a travel preparation list, and a journal included in the application.

2.4 Research Gap

The academic literature review shows that there is a substantial development in using artificial intelligence, machine learning, and recommender systems for travel planning. The study of AI-based trip planners identifies the enhancement of itinerary optimization, personalization, and efficiency to a single user. Research on large language models demonstrates that they are able to help in idea generation and information organization during collaborative planning exercises, and group recommender systems bring in new methods of dealing with preference conflicts and maximizing group satisfaction. Moreover, the results of optimization models and cost prediction methods in similar fields prove the possibility of intelligent systems handling complex and non-linear decision variables. Nevertheless, these works are mostly focused on single points of travel planning, including personal optimization, static group recommendation, or support of AI, whereas they do not have a comprehensive and end-to-end collaborative planning process.

There is also a distinct gap in academic literature on the question of practical implementation of the real time collaboration, organization of group decision-making, natural language interaction as well as predictive cost estimation in one travel planning system. The present body of research typically presupposes controlled settings, fixed limitations, or limited interaction between the user, which limits the possibility of its application in the real-life group travel context where preferences change over time and decisions have to be negotiated in an open manner. This gap needs to be filled so as to bridge the gap between the hypothetical optimization models and the real requirements of groups making trip plans jointly. The gap in this research case study is that it proposes and implements a joint web-based travel planning system, which combines collaborative planning areas, formalized voting systems, NLP-enhanced communications, contextual travel information and predictive cost modelling to aid group decision-making dynamically and transparently.

Itinerary aggregation, route discovery and expense tracking Commercial travel planning systems like TripIt, RoadTrippers and SaveTrip have useful functionality. These tools are highly usable and efficient for individual travelers, and they have little sharing features or asynchronous collaboration. Nevertheless, they are missing real-time, multi-user collaboration, structured, consensus-building, tools, built-in cost prediction among group members, and facilitative group preferences. Consequently, users still use outside communication tools and manual coordination in the process of planning trips as a group which results in the fragmented workflow and less efficient planning. This is a gap that cannot be ignored since the commercial adoption emphasizes real demand of the user, but the current systems are not able to complete the collaborative group planning. The proposed research will address this gap by creating a platform that has the practical advantages of the current platforms but with more collaborative functionalities and features so that the groups can plan, assess and finalize the trips together in one platform that is interactive and user-friendly.

Table 1: Comparison between existing systems and TripMate

System / Feature	TripIt	RoadTrippers	SaveTrip	TripMate
Itinerary Creation	From emails and fixed templates	Fixed templates	Fixed templates	Customizable itineraries with structured and rich formatting
Real-time Collaboration	No	No	No	Yes
Itinerary Sharing	Collaboration (view-only)	Collaboration (limited editing)	Export as PDF	Full collaborative editing
Budgeting Tools	No	No	Multi-currency expense tracking	Integrated cost estimation, tracking, and prediction
Document Storage	No	No	Yes	Yes
Route Planning	No	Interactive maps	Via Google Maps integration	Integrated route planning
Navigation	No	Turn-by-turn directions	No	No
Travel Guides	No	Curated guides	No	Context-aware recommendations
Checklists	No	No	Basic checklists	Customizable group checklists
Reminders	Booking alerts	Disruption alerts	No	Custom reminders and notifications
Platforms	iOS, Android	Web, iOS, Android	iOS, Android	Web, iOS & Android (Future)

The above Table 01 shows the comparison between the commercially available systems TripIt, RoadTrippers, SaveIt against TripMate.

2.5 Market Research

In this section the market research is given to obtain insights into user behaviours, expectations and challenges associated with group travel planning. It examines primary data gathered by means of surveys to determine popular planning habits, pain points, and desired features of a collaborative travel planning app. The problem identified in the literature review is tested with the findings of this section and applied to functional and non-functional requirements as well as to assure that the design of the TripMate system is based on the actual user needs and market demands.

2.5.1 Data Gathering

The proposed system has gathered data on functions that influence their features, individual preferences, functions, user experience and other functions. Market research was conducted through an online survey. The survey is presented in the appendix of this file.

2.5.2 Requirement Analysis

What is your age group?

22 responses

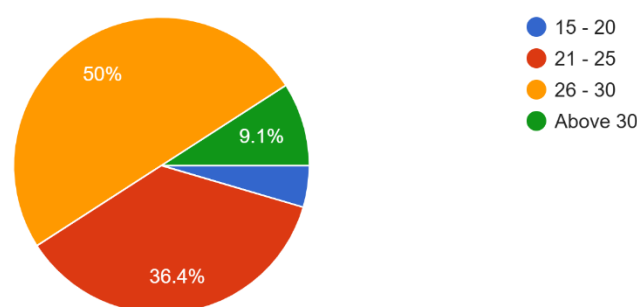


Figure 12 : Survey questionnaire, what is your age group?

The above pie chart indicates that the user population of the app is significantly young, as 86.4% of 22 total respondents are aged between 21-30. The heaviest segment is 26–30, accounting for exactly 50.0% of all responses, followed by the 21–25 at 36.4%. The rest of the subjects consist of the Above 30 group (9.1%) and 15-20 group (4.5%).

How often do you travel within Sri Lanka?

22 responses

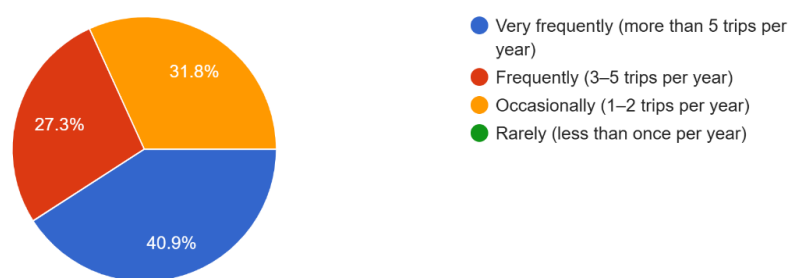


Figure 13 : Survey questionnaire, how often do you travel in Sri Lanka?

The above pie chart reveals the users are very active travelers, with a large majority traveling multiple times per year. Two-thirds of respondents (68.2%) travel regularly: 40.9% (9/22) "Very frequently (more than 5 trips per year)" and 27.3% (6/22) "Frequently (3–5 trips per year)". The other fraction, 31.8% (7 of 22), goes "Occasionally (1–2 times per year)". Such a high volume of travel also indicates that the target group faces group planning issues frequently, which would in turn suggest likely strong demand for an effective trip planning system such as TripMate.

How do you usually plan your trips?

22 responses

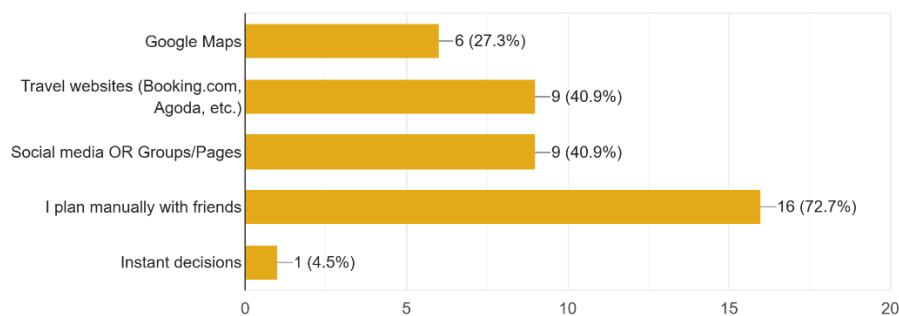


Figure 14 : Survey questionnaire, how do you usually plan trips?

This bar chart on "How do you usually plan your trips?" strongly indicates that the respondents seriously depend on manual planning approaches. Manual planning with friends though is currently the most preferred approach and shared by 72.7 % (16 out of 22) as reported in all respondents. Other popular non-specialist tools include Travel sites (Booking. com, Agoda etc.)” and “Social media OR Groups/Pages” by 40.9% (9 mentions) of the sample. Also, Google Maps acts as an organizational tool for 27.3% (6 mentions) of respondents. None, except for a tiny minority of 4.5% (1 occurrence), rely on Instant decisions. This information highlights that the initial activity is not automated nor done individually but in collaboration indicating a gap for a devoted and feature-rich collaborative platform.

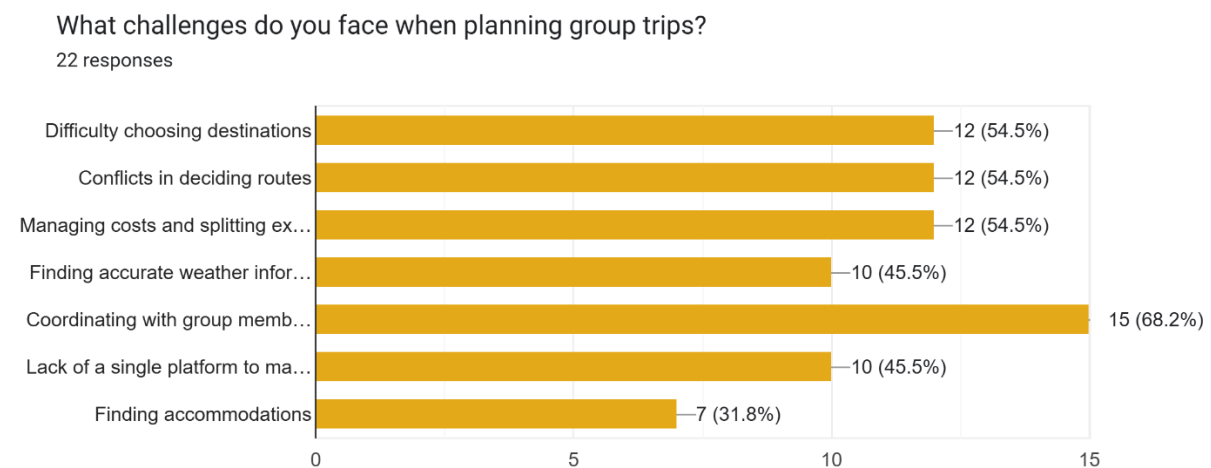


Figure 15 : Survey questionnaire, what challenges do you face when planning group trips?

This bar chart shows the main obstacle faced by the respondents was "coordinating with group members", as 68.2% of the survey participants, out of a total of 22, mentioned this issue when planning a group trip. The next most common difficulties encountered by the group were a series of issues which were related to decision-making and costs, these being the difficulty in choosing the destination, disputes in deciding the route, and expenses management and splitting up expenses were each mentioned by 54.5% of the respondents.

Half the number of the respondents, both “unreliable weather forecasts” and “absence of a one-size-fits-all platform to control all the aspects of the trip” were, each cited by 45.5% or ten of the group. One of the physical logistics issues, finding accommodations, was a challenge for a fairly significant 31.8% of the respondents, and it's clear from the data that the main problems lie in group coordination, group decision-making, and financial planning, and that a well-designed application should focus on these areas.

Have you ever used a platform specifically designed for group trip planning?

22 responses

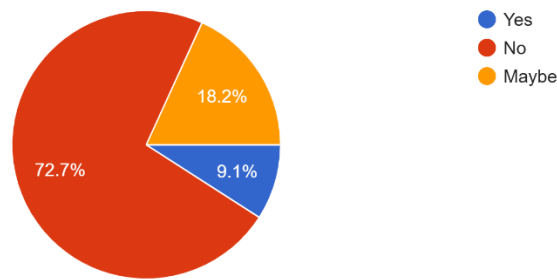


Figure 16 : Survey questionnaire, have you ever used a platform specifically designed for group trip planning?

How useful would a webapp be that allows you to plan trips collaboratively with friends?

22 responses

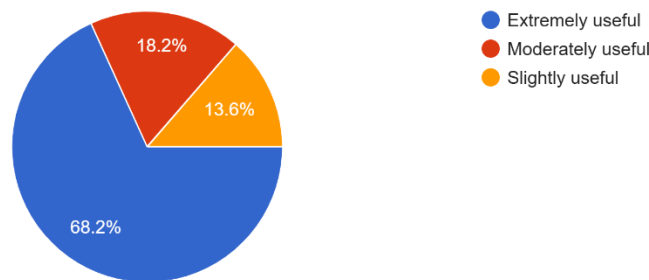


Figure 17 : Survey questionnaire, how useful would a web app be that allows you to plan trips collaboratively with friends?

The above pie charts (figures 16 and 17) show analysis of two survey data indicates a clear need for a collaborative planning tool, despite the current low usage of similar platforms. Most respondents, 68.2% (15 out of 22), found a web application for collaborative trip planning Extremely useful, and another 18.2% (4 mentions) considered it Moderately useful. This suggests that a web app designed for this purpose would be well received. However, 72.7% of respondents (16 out of 22) have never used a dedicated platform for collaborative planning. Only 9.1% (2 mentions) had experience with such a platform, and 18.2% (4 mentions) were unsure. This difference points to a situation where, even though there is a strong desire for specialized collaborative planning solutions, existing options have not connected with or met the needs of their intended users.

How important are the following features to you?

22 responses

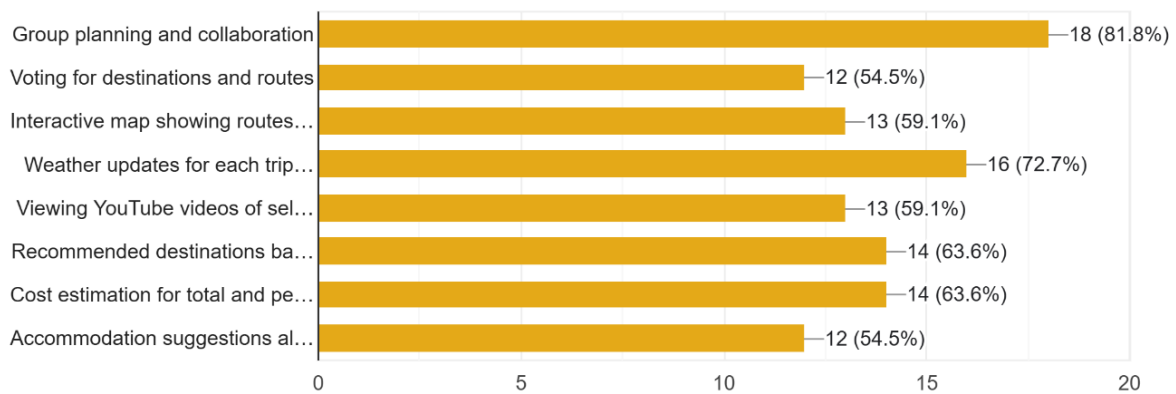


Figure 18 : Survey questionnaire how important are following features to you?

This bar chart results show a clear demand for an app that helps groups plan things together. Most people, 81.8% (18 out of 22), picked Group planning and collaboration as the main feature they want, which fits with the idea of a group planning app. Users also voted for features that solve common problems when planning, like Weather updates for each trip day (72.7% or 16 mentions), Recommended destinations based on your search (63.6% or 14 mentions), and Cost estimation for total and per-person expenses (also 63.6% or 14 mentions). On top of that, Users are interested in features that are more interactive and include media, such as Interactive map showing routes and travel spots (59.1% or 13 mentions), Viewing YouTube videos of selected locations (59.1% or 13 mentions), and Voting for destinations and routes (54.5% or 12 mentions), all chosen by more than half of those surveyed.

Would you use a feature where the system suggests travel destinations based on your input? (Example: "best beaches for camping")

22 responses

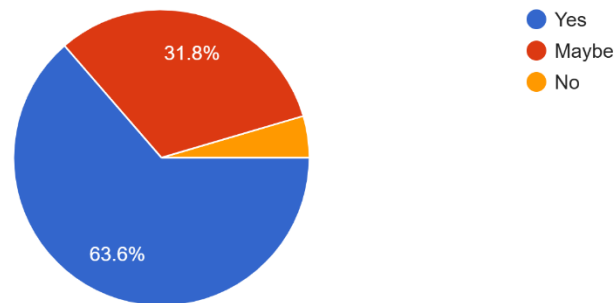


Figure 19 : Survey questionnaire, would you use a feature where the system suggests travel destinations based on your input?

The above pie chart shows responses to a survey questionnaire on system features offering travel destination suggestions driven by user input. Most respondents, 63.6% (14 out of 22), affirmed they would use such a feature. This strong positive response indicates users are looking for help with the frequently mentioned difficulty in picking destinations. A good number, 27.3% (7 mentions), replied 'Maybe', which implies they are open to the feature, but their choice could rest on how accurate and good the system's suggestions are. Just 9.1% (1 mention) of those asked outright said 'No' to using the suggestion tool.

How valuable is an automated cost estimation tool when planning a trip?

22 responses

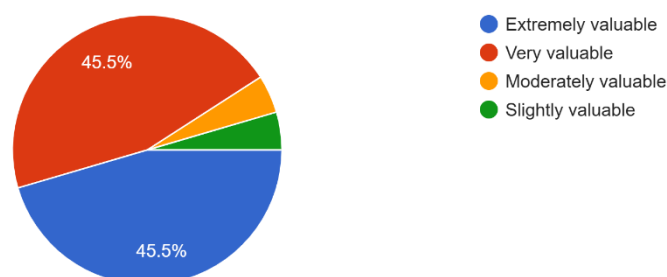


Figure 20 : Survey questionnaire, how valuable is an automated cost estimation tool when planning a trip?

The above pie chart shows analysis of the data reveals a strong inclination among respondents for an automated cost estimation tool in trip planning. A substantial 77.3% (17 out of 22) of participants considered this feature either extremely valuable (45.5% or 10 mentions) or very valuable (31.8% or 7 mentions). This shows the idea that managing group finances and dividing expenses, often seen as a problem, represents a crucial application for the app. The remaining respondents showed moderate interest, with 13.6% (3 mentions) classifying it as Moderately valuable and only 4.5% (1 mention) as Slightly valuable. This high value suggests that including an accurate cost estimation feature would be vital for TripMate to do well.

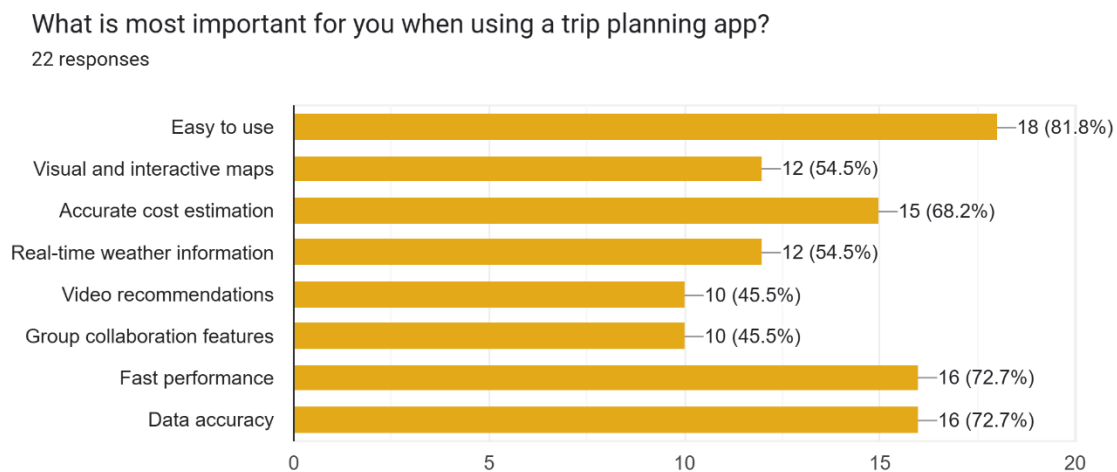


Figure 21: Survey questionnaire, what is most important for you when using a trip planning app?

The above bar graph shows when users pick a trip planning app, they care most about how easy it is to use and how well it works. A huge majority, 81.8% (18 out of 22), said Easy to use was the most important thing. Right behind that, both Fast performance and Data accuracy were chosen by 72.7% (16 people each), showing that speed and trustworthy info matter a lot. Accurate cost estimation was also seen as key by 68.2% (15 people), which is helpful for budgeting trips with others. Things that make planning nicer, like Visual and interactive maps and Real-time weather information, were each picked by 54.5% (12 people) as most important. This tells us that users want apps that are easy to use, have good info, and offer helpful planning tools.

2.5.3 Conclusion

The market research evidence is a solid empirical evidence of the risk of establishing TripMate as a collaborative group traveling planning platform. According to the results of the survey, the main target audience is represented mainly by young adults between the age of 21-30 years, as they are frequent travelers, thus frequently face the problems related to group trip planning. Such demographic profile indicates a technologically conscious group of users whose chances of using digital solutions to enhance coordination, efficiency, and decision-making are high.

The figures show clearly that the existing practices of trip planning are mostly manual and disjointed. Majority of the respondents do not use collaborative planning applications but informal coordination with friends, generic travel websites, social media sites and mapping applications. This dependency on detached tools shows the absence of characterized solutions that can be used to facilitate coordinated decision-making in groups. The fact that the use of dedicated collaborative planning platforms before was minimal, even though the perceived usefulness is high, may also suggest that the current systems do not meet the expectations and practical demands of the users.

The issue of group coordination became the greatest difficulty encountered during the process of planning the trip, and the issue of collective decision-making, cost management, or expense sharing came next. Most of the respondents identified these problems, which supports the conclusion that group travel planning is not just an individual optimization problem but a complicated social and financial coordination challenge. Other issues like poor weather forecasting and the lack of a centralized platform are also other issues that justify the need to have a complete, one platform collaborative system.

The analysis of the features preferences shows that it is strongly demanded to work in a group in real-time, recommend destinations smartly, estimate the cost automatically, and provide contextual information like the weather forecast. The perceived value of automated cost estimation and expense management is high which means that financial transparency is a highly important element that helped to avoid conflict and enhance trust between group members. Moreover, interactive maps, voting systems, and multimedia content are of great interest, which implies that the users are willing to see the tools of modern travel planning to be functional and engaging at the same time.

Lastly, the most influential issues in choosing a trip planning application were found to be usability, performance, and precision of the data. This is where the fact that there should be not only a rich system but also an intuitive, responsive, and reliable one comes to play. Together, these data prove that there is an evident gap in the market of the user-centered, collaborative web-based travel planning platform. TripMate can easily fill this gap by adding real-time working together, smart suggestions, price anticipation, and a simplified user experience to one unified system that would meet the demands of the present-day group travelers.

3. Project Plan

This section presents the TripMate development project plan, detailing the project's structure, organization, and timeline to accomplish its goals. The project plan specifies the major stages of the work, the order of the activities, and the various planning documents that are used to track progress and assure delivery on time. By providing a well, defined itinerary, this section serves as a window into how the project progresses from the preliminary planning and data analysis to the system development, testing, and final rollout. The main planning instruments covered are the project overview, the Work Breakdown Structure (WBS), and the Gantt chart.

3.1 Overview of the project

The TripMate project revolves around the creation of a collaborative web, based travel planning system that is a group coordination, decision, making, cost estimation, and itinerary management problem solver. The project intends to combine the features of shared trip planning, real time collaboration, intelligent destination suggestions, weather awareness, and automated cost estimation in one, easy, to use platform.

The project is carried out in stages, starting with planning and data analysis to ensure that system requirements are based on user needs and research findings. This first phase of work also involves defining requirements, analyzing the survey, and preparing data for machine learning components. The development work is then split into two major implementation phases. The first phase focuses on core system functionality and backend services, while the second one is about user interface development, system integration, and deployment.

During the entire project lifecycle, the main idea of going forward step by, step, validating all the time and being in line with the expectations of users identified through market research is kept. Such a well, organized though adaptable method is a guarantee that TripMate will turn into a handy, extendable, and efficient tool for collaborative group travel planning.

3.2 Gantt Chart

The Gantt chart illustrates the timeline of the TripMate project and visually represents the duration, sequencing, and dependency of activities defined in the WBS. As shown in the chart, early project phases such as planning and data analysis are completed prior to development activities, ensuring a strong foundation for system implementation.

The chart highlights two primary development periods aligned with the project sprints. Sprint 1 focuses on backend and architectural components, while Sprint 2 emphasizes frontend

development, integration, and deployment activities. Milestones such as testing completion and project launch are clearly identified, enabling effective tracking of progress and deadlines.

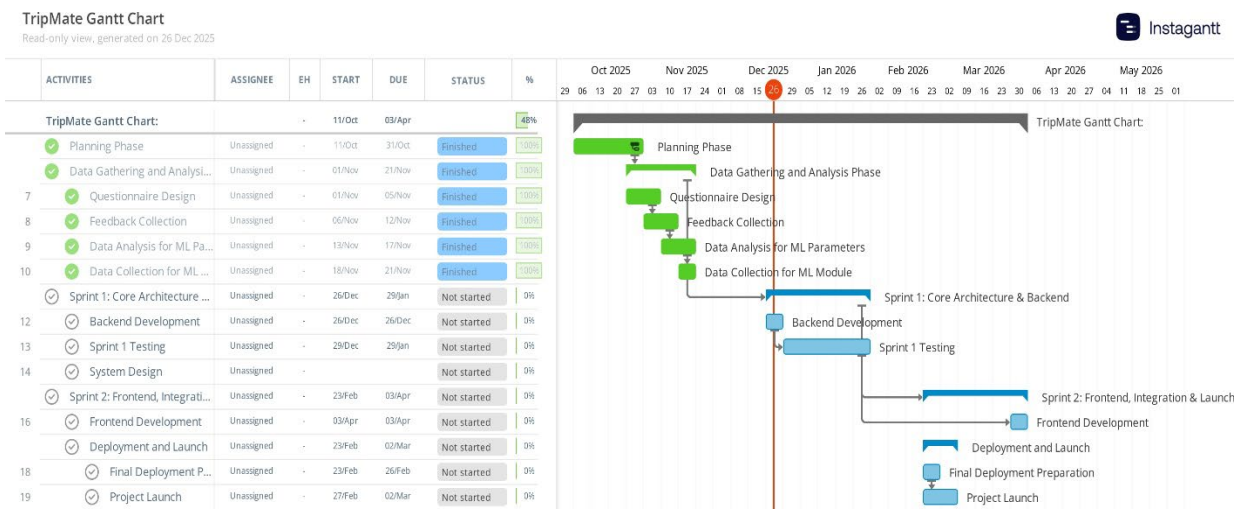


Figure 22 : TripMate Gantt Chart

The above figure shows the Gantt Chart of TripMate. The task in green shows the completed tasks while in blue shows the unfinished tasks.

3.3 Work Breakdown Structure

The Work Breakdown Structure (WBS) is used to break down the TripMate project into logical stages and manageable subtasks. It illustrates in a hierarchy all the work needed to achieve the project and acts as a basis for the timing, checking, and controlling of the project activities. As the WBS defines the deliverables and the tasks linked to them explicitly, it enhances the planning accuracy and is a great tool for the smooth working of the project.

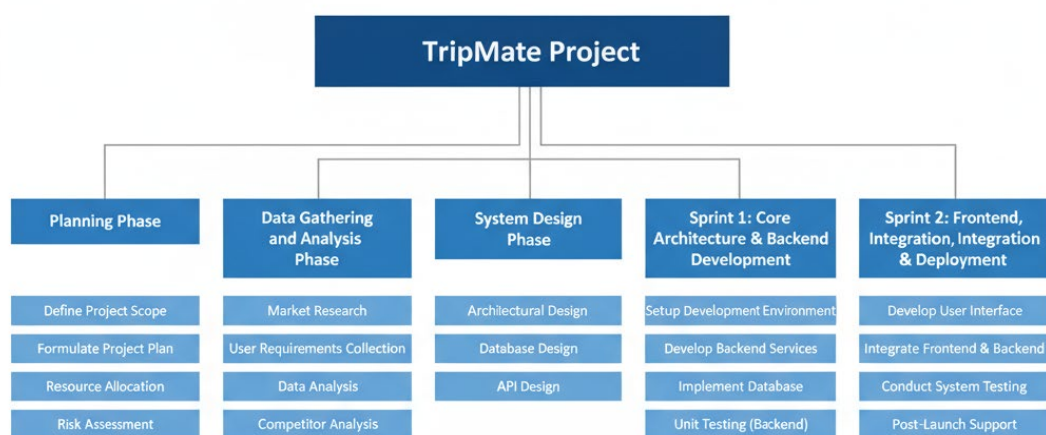


Figure 23 : WBS

Figure 23 shows the WBS of the TripMate project.

4. Planning the Artifact

This section provides details of the systematic approach that would be used in planning, designing, implementing and evaluation of the TripMate artifact. It justifies the choice of the development methodology, overview of the previous planning activities and details the requirement analysis, design and implementation strategy, testing plan and risk management. This section aims at illustrating how the project is converted out of conceptual knowledge into an actual practical working system, via organized planning and engineering.

4.1 Methodology

The TripMate software development is based on the Agile software development methodology, which is applied using two iterative sprints. Agile was chosen because of the exploratory and dynamic character of the project, which includes the incorporation of collaborative features, AI-assisted information planning, and cost estimation systems with references to the constant user feedback and research results. The nature of group travel planning systems is characterized by the evolution of the user requirements, clash of preferences and the systematic enhancement of the features of the system like collaboration, voting and budgetary tools. Agile is very favorable to incremental development, early validation, and ongoing enhancement and is thus suitable to a user-based feature-based application in web-based applications such as TripMate. The sprint-based methodology enables development of the core functionalities, testing and refining of the same, and subsequently proceeds to the development of higher-level functionality and system integration.

Table 2 : Advantages and Disadvantages of Agile methodology

Advantages of Agile	Disadvantages of Agile
Supports iterative development and flexibility	Requires strong project discipline
Enables continuous feedback and improvement	Scope creep may occur if not controlled
Early identification of issues through frequent testing	Documentation may be less detailed
Well-suited for user-centered systems	Requires active stakeholder involvement
Facilitates faster delivery of functional components	Difficult to estimate timelines precisely

4.2 Planning the Artifact

Artifact planning requires the choice of a proper development methodology, the definition of system requirements, the description of the design and implementation strategy, and the definition of testing, evaluation, and risk management strategies. The focus is made on the artifact that is to directly address the challenges that were identified in collaborative group travel planning such as coordination, decision-making, cost estimation, and usability. This section will show how theoretical understanding, market research and technical factors are integrated systematically to drive the process of designing a strong, user friendly and scalable collaborative travel planning system.

4.3 Requirement gathering and Analysis

The hardware, software, functional and nonfunctional requirements of TripMate are listed below.

Hardware Requirements

Table 3 : Hardware Requirements needed for TripMate

Requirement	Description
Mobile device	Smartphone capable of running modern web or mobile applications (Android or iOS)
PC / Laptop (for development)	Minimum 8GB RAM, Intel i5 processor or equivalent, SSD storage for efficient development
Server infrastructure	Cloud-based server for hosting backend services and APIs
Internet connection	Stable internet connection for real-time collaboration, API access, and data synchronization

The above table shows the hardware requirements needed for TripMate.

Software Requirements:

Table 4 : Software Requirements for TripMate

Requirement	Software / Framework
Backend framework	ASP.NET Core / Node.js
Frontend framework	React JS / React Native
Database	SQL / MySQL
Machine learning	Python-based ML libraries (e.g., TensorFlow, PyTorch for cost prediction models)
Natural Language Processing	NLP libraries for analysis and recommendation generation
APIs	Google Maps API, Weather API
Version control	Git / GitHub
Cloud hosting	AWS / Azure

The above table shows the software requirements needed for TripMate.

Functional Requirements:

Table 5 : Functional Requirements of TripMate

Requirement	Description
User authentication	Allow users to register, log in, and manage profiles
Group creation and management	Enable users to create groups and invite members
Collaborative itinerary creation	Allow multiple users to create and edit itineraries

Requirement	Description
Real-time collaboration	Synchronize changes across group members in real time
Voting mechanisms	Support voting for destinations, routes, and activities
Cost estimation	Automatically estimate total and per-person travel costs
Expense tracking	Track and split expenses among group members
AI-based recommendations	Suggest destinations and activities based on group preferences
Weather integration	Display weather forecasts for selected trip dates
Itinerary sharing	Enable sharing within and outside the platform

The above table shows the functional requirements of TripMate.

Non-functional requirements:

Table 6 : Non-functional Requirements of TripMate

Requirement	Description
Cross-platform support	System should operate on web and mobile platforms
Scalability	System should support increasing users and group sizes
Performance	Fast response time for collaboration and data updates
Usability	Interface should be intuitive and easy to use
Data accuracy	Travel data, costs, and recommendations must be reliable
Security	Protect user data, group information, and financial details
Availability	System should be accessible with minimal downtime
Maintainability	Codebase should allow future enhancements and integrations

The above table shows the non-functional requirements of TripMate.

4.4 Design and Implementation

TripMate was designed and implemented in a layered architecture which is modular. To enhance scalability and maintainability, the system will have a strict division among frontend, backend and data layers. The frontend is dedicated to the user interface, interfaces with the collaboration, and visual planning modules, whereas the backend is the one that deals with business logic, user coordination, cost estimation, and AI-aided capabilities. The implementation is done in small sprints where essential functions are tested first before any sophisticated functions are implemented. This strategy helps in the ongoing improvement and lessening of technical risk.

4.5 Testing and Evaluation

The test process is done during the lifecycle development of the system to guarantee reliability, accuracy and usability of the system. Unit testing checks all the elements, and integration testing is to guarantee the functionality of the system modules. The user acceptance testing is done to determine the appropriateness of the system to the expectations of the users as they are perceived during the market research. The assessments are geared towards usability, performance and the effectiveness of features especially in facilitating the coordination of a group of people, decision making and cost management.

4.6 Risks and Risk Mitigation Techniques

Some of the factors that may pose risks to the project are technical complexity, integration issues, and scope creep. Such risks are addressed with incremental development, early testing, third party API that is reliable, and scope definition in sprints. Frequent review of progress and updating of documentation also help to minimize uncertainty on a project and make sure that TripMate is not deviating off the track of its goals and deliverables.

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Appendix A – Questionnaire

Questions

Responses 22

Settings



Section 1 of 5

TripMate – A Collaborative Trip Planning Webapp

TripMate 🌐 ✨ is a web app that helps groups plan trips using maps, weather updates, videos, and cost estimates. This survey gathers your opinions to help design the app and improve its features 🥰 .

Email *

Valid email

This form is collecting emails. [Change settings](#)

Section 1: Participant Information

Description (optional)

What is your age group?

Multiple choice

☐ 15 - 20

☐ 21 - 25

☐ 26 - 30

☐ Above 30

☐ Add option or [add "Other"](#)

Required

What is your gender?

☐ Male

☐ Female

☐ Other

☐ Prefer not to say

How often do you travel within Sri Lanka?

☐ Very frequently (more than 5 trips per year)

☐ Frequently (3–5 trips per year)

☐ Occasionally (1–2 trips per year)

☐ Rarely (less than once per year)

Section 2: Current Travel Planning Behaviour



Description (optional)

How do you usually plan your trips?

- ☐ Google Maps
- ☐ Travel websites (Booking.com, Agoda, etc.)
- ☐ Social media OR Groups/Pages
- ☐ I plan manually with friends
- ☐ Other:

What challenges do you face when planning group trips?

- ☐ Difficulty choosing destinations
- ☐ Conflicts in deciding routes
- ☐ Managing costs and splitting expenses
- ☐ Finding accurate weather information
- ☐ Coordinating with group members
- ☐ Lack of a single platform to manage everything
- ☐ Finding accommodations

Have you ever used a platform specifically designed for group trip planning?

- ☐ Yes
- ☐ No
- ☐ Maybe

Section 3: TripMate Feature Evaluation

Description (optional)

How useful would a webapp be that allows you to plan trips collaboratively with friends?

- ☐ Extremely useful
- ☐ Moderately useful
- ☐ Slightly useful

How important are the following features to you?

- ☐ Group planning and collaboration
- ☐ Voting for destinations and routes
- ☐ Interactive map showing routes and travel spots
- ☐ Weather updates for each trip day
- ☐ Viewing YouTube videos of selected locations
- ☐ Recommended destinations based on your search
- ☐ Cost estimation for total and per-person expenses
- ☐ Accommodation suggestions along the route

Would you use a feature where the system suggests travel destinations based on your input?
(Example: "best beaches for camping")

- ☐ Yes
- ☐ Maybe
- ☐ No

How valuable is an automated cost estimation tool when planning a trip?

- ☐ Extremely valuable
- ☐ Very valuable
- ☐ Moderately valuable
- ☐ Slightly valuable

Section 4 of 5

Section 4: User Expectations



Description (optional)

What device would you most likely use TripMate on?

- ☐ Mobile phone
- ☐ Tablet
- ☐ Laptop/PC

What is most important for you when using a trip planning app?

- ☐ Easy to use
- ☐ Visual and interactive maps
- ☐ Accurate cost estimation
- ☐ Real-time weather information
- ☐ Video recommendations
- ☐ Group collaboration features
- ☐ Fast performance
- ☐ Data accuracy

Would you be interested in using TripMate once it is developed?

- ☐ Yes
- ☐ No
- ☐ Not sure

After section 4 Continue to next section



Section 5 of 5

Section 5: Open Feedback



Description (optional)

What features would you like to see added to TripMate?

Short answer text

Do you have any concerns about using a group-based trip planning webapp?

Short answer text

Any additional suggestions or comments?

Long answer text

Appendix B – Work Breakdown Structure

Work Breakdown Structure for TripMate

TripMate Project

- 1. Planning Phase**
 - Project scope definition
 - Requirement identification
 - Project planning and documentation
- 2. Data Gathering and Analysis Phase**
 - Questionnaire design
 - Feedback collection
 - Data analysis for machine learning parameters
 - Data collection for ML module
- 3. System Design Phase**
 - Overall system architecture design
 - Database design
 - API and integration design
- 4. Sprint 1: Core Architecture & Backend Development**
 - Backend development
 - Database implementation
 - API development
 - Sprint 1 testing
- 5. Sprint 2: Frontend, Integration & Deployment**
 - Frontend development
 - Backend–frontend integration
 - System testing and validation
 - Deployment and launch preparation
 - Final deployment and project launch